

Introduction to System Analysis and Design

Systems

- A *system* is a set of components that interact to accomplish some purpose
 - E.g. a nervous system, an economic system, a language.
- A business is also a system
 - Components of a business system: research, manufacturing, marketing, sales etc.
- Key elements:
 - Input, Output, Environment, Boundaries, Processor (Operational component), Control (Decisional component), Feedback

Example: Online Food Delivery System

● **Suppose Alice orders a pizza using a food delivery app.**

1. Input: The data or resources entered into the system. **Ex:** Customer's order, Delivery address, Payment information

Scenario: Alice selects a pizza, enters her address, and pays online.

2. Processor (Operational Component): The part of the system that processes the input. **Ex:** The restaurant receives the order, The pizza is prepared, A delivery rider is assigned, The rider picks up the order.

Scenario: The restaurant prepares the pizza, and the system assigns the nearest delivery rider.

3. Output: The final result produced by the system. **Ex:** The customer receives the pizza, The order is marked as delivered.

Scenario: Alice receives her pizza successfully.

Example: Online Food Delivery System

4. Environment: External factors that affect the system but are not part of it. **Ex:** Traffic, Weather, Internet connection.

Scenario: Heavy rain causes a delay in the delivery.

5. Boundary: The boundary defines what is inside and outside the system.

Inside the system:

- Food delivery app
- Restaurant
- Delivery rider
- Payment gateway

Outside the system:

- Weather
- Traffic
- Customer's personal life

Scenario: The app, restaurant, and rider are part of the system, while the weather is outside the system.

Example: Online Food Delivery System

6. Control (Decisional Component): The component that makes decisions and manages the system. **Ex:** Selecting the nearest rider, Verifying payment, Accepting or rejecting an order.

Scenario: The system automatically chooses the nearest available rider to deliver the pizza.

7. Feedback: Information returned after the output to improve the system. **Ex:** Customer rating, Customer review, Complaint

Scenario: After receiving the pizza, Alice gives the service **5 stars** and writes, "The food was delicious and delivered on time."

Systems

- Some systems interact with their environments to achieve their purpose; they are ***open systems***
 - All real life systems are open systems (University, Hospital, Bank)
- Those which do not, are ***closed systems***
 - Hypothetical (Sealed Thermos Flask)
- Indispensable element of a system: the ***control element***
- The control element compares the performance of the system against standards and makes adjustments if performance deviation is too far from the standards

Systems: The control element

- The control model consists of:
 - A standard
 - Performance measurement metrics
 - Performance comparison techniques
 - *Feedback*
- The results of comparison are *fed-back* to the system input, for adjustment

Systems: The control element

- Components of a system may themselves be smaller systems or subsystems
- **Main System:** Hospital

Subsystems:

- Reception System
- Pharmacy System
- Laboratory System
- Billing System
- Patient Record System

System Models

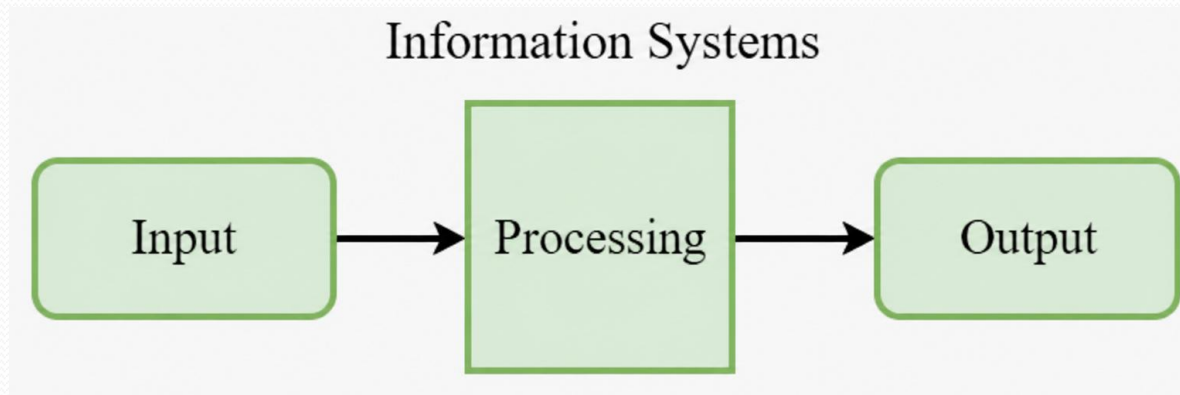
- Representation of reality with which the system is concerned
- Enables visualization of important aspects of the system under study, by the system analyst:
 - Its key elements and their interrelationships
- Types:
 - **Schematic Models:** 2-D charts depicting system elements and their linkages
 - **Flow System Models:** Depicts flow of materials and data
 - **Static System Models:** 2-D plots exhibiting one pair of relationships, e.g. time-activity (GNATT chart)
 - **Dynamic System Models:** Used to represent a dynamically changing system, e.g. a business system

Business Systems

- Every business system has the features of a system in general
- Their purpose is to produce goods, products or technology that fulfill a demand in market
- A business systems interacts with its environment to acquire resources – materials, manpower, knowledge
- Performance of a business organization is continually evaluated and compared against market standards by managers
- Adjustments made by employees to meet the standards

Information Systems

- The most indispensable component of any present day business organization is its *information system*
- Information system is that abstract component of a business system which deals with flow of information or valuable data from one entity to another
 - It processes input, maintains data and produces output



- Subsystems include: hardware, software, data storage

Users of Information Systems

- Broadly, the managers and employees of an organization who interact with the information system
- Specifically,
 - **Hands-on users:** Users interacting with the systems through terminals, providing input and collecting output
 - **Indirect users:** Users benefitting from the output (results and reports); no direct interaction with the system
 - **User managers:** Have organizational responsibility to control system's effectiveness, by overseeing resource utilization
 - **Senior management:** Monitors risks from information system failure; strategically plans system development

Challenges for IS Developers

- Developing an IS
 - is all about channeling human knowledge and ideas to harness a machine to perform specific task(s)
- An IS should
 - function correctly
 - be easy to use
 - Help people work more efficiently
- The technology developed should not be an obstacle even for the most novice user, to get the desired results

System Analysis

- The process of investigating the present status and future requirements of a (business) system with the target of improving its performance through improvisation or innovation.

Design of a Business Systems

- To plan a new business system or to replace or complement an old system, eliminate its drawbacks.
- For system design, one should have a prior knowledge of the system requirements. This includes
 - Thorough understanding of the old system's operations or the user's requirements
 - A plan for design and development of a sufficiently effective system
- This gives rise to the need for system analysis

System Analysis

- Includes:
 - Collection of facts (possibly from users)
 - Interpretation of facts
 - Investigating problems and requirements
 - Determining costs and benefits
 - Solution optimization
 - Producing reports recommending solutions and improvements

System Analysis

Example: University Result Management System

- **Collection of facts :** Collect students' names, IDs, and marks.
- **Interpretation of facts:** Calculate GPA, assign grades, and determine Pass/Fail status.
- **Investigating problems and requirements:** Identify issues in the current manual system (e.g., calculation errors and delays) and gather user requirements.
- **Determining costs and benefits:** Evaluate the cost of developing the new system and the benefits, such as faster processing and fewer errors.
- **Solution optimization:** Design a system that is accurate, efficient, and easy to use.
- **Producing reports recommending solutions and improvements:** Prepare a report recommending an online Result Management System to improve efficiency and reduce manual work.

System Analysis

- Does NOT include:
 - Investigating automation opportunities
 - Altering existing technologies
 - Determining the technically best solution
 - Rather determining the solution which best benefits the business
- Analysis specifies *what* the system should perform
- Design specifies *how* to achieve the goal

Role of System Analysts

- Various roles of a System Analyst:
 - System Analysis:
 - Collecting relevant facts and information through system studies. Hence determining requirements
 - System Analysis and Design:
 - Complete system study plus design specification development
 - Work on fewer projects
 - System Analysis, Design and Programming:
 - System investigation, design specification development, implementation of the design
- Smaller firms assign more roles to a system analyst
- Larger firms hire people who specialize in specific roles

Categories of IS

- Transaction Processing System (TPS)
- Management Information System (MIS)
- Decision Support System (DSS)

Transaction Processing Systems(TPS)

- A transaction processing system (TPS) is a type of **data management** information-processing software used during a business transaction to manage the collection and retrieval of both customer and business data.

Transaction Processing Systems(TPS)

●Functions:

1. **System runtime functions:** Basic functions associated with the purpose of processing a transaction while maintaining data integrity, availability and security, all with fast response times and high transaction throughput.
1. **System administration functions:** Functions associated with system administration, such as the configuration, monitoring and management of the TPS.

Transaction Processing Systems(TPS)

●Functions:

3. Application development functions: To better suit the particular business application, a modern TPS offers customization features to access data, perform intercomputer communications and design and manage unique user interfaces.

Ex-1: A university develops a TPS for student registration. Using application development functions, the university customizes the system by allowing students to register courses, teachers to enter marks, administrators to manage student records, and the system automatically shares data with the finance and library systems. This makes the TPS suitable for the university's specific needs.

Transaction Processing Systems(TPS)

●Functions:

Ex-2: A hospital customizes its TPS according to its needs. Receptionists can register patients and access patient records, doctors can view medical history and write prescriptions, laboratory staff can upload test reports, pharmacists can view prescriptions, and the billing system automatically calculates the patient's bill. Each user has a separate interface, and all departments communicate with each other through the TPS.

TPS

●Components:

1. **Input:** Any number of transactions, including invoices, bills, coupons and other types of orders like a purchase order, may be treated as inputs in a TPS. Theoretically, any type of order entry can be considered input data.
2. **Output:** A TPS can generate a variety of use-case-specific outputs ranging from cash flow reports to receipts, and it can be utilized for record-keeping, data analysis, tax reporting and other official business purposes.
3. **Processing system:** The processing system of a TPS reads the input, completes any data modifications or updates, and creates a useful output, such as a confirmation of sale or inventory report.

TPS

●Components:

4. **Storage:** While storage may, in some cases, refer to physical data storage hardware, an average TPS will also create easily navigable directories for storing both input and output data, typically in some form of database.

TPS

●Components:

Example: Hospital Management System (TPS)

- **Input:** A patient arrives at the hospital and provides their **name, age, symptoms, and appointment details**.
- **Processing:** The TPS verifies the patient information, creates or updates the patient record, assigns a doctor, and calculates the consultation fee.
- **Output:** The system generates a **patient registration slip, appointment confirmation, and bill**.
- **Storage:** Finally, the patient's details, appointment information, prescription, and billing records are **stored in the hospital database** for future use.

Flow:

Patient Information (Input) → System Processes the Data → Registration Slip & Bill (Output) → Records Saved in Database (Storage)